Attention Walkthrough: Feature Mode

This walkthrough illustrates how the learned attention scores enable the model to identify the context item with the closest feature to the query.

Step 1: Learned Scores Matrix

After training on the feature-identification task, the model learns a banded scores matrix S where nearby features have higher scores:

$$S = \begin{bmatrix} 4 & 3 & 1 & -1 & -3 \\ 3 & 4 & 3 & 1 & -1 \\ 1 & 3 & 4 & 3 & 1 \\ -1 & 1 & 3 & 4 & 3 \\ -3 & -1 & 1 & 3 & 4 \end{bmatrix}$$

Diagonal: highest (same feature)
Off-diagonal: decreasing with distance

Step 2: Query and Context

Query (reference): feature index 2

$$X_q = \begin{bmatrix} 0 & 0 & \mathbf{1} & 0 & 0 \end{bmatrix}$$

Context: 5 items with features $[0, 4, 0, 4, 1]$

$$X_c = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$\leftarrow$ pos 0: feature 0, dist=2
$\leftarrow$ pos 1: feature 4, dist=2
$\leftarrow$ pos 2: feature 0, dist=2
$\leftarrow$ pos 3: feature 4, dist=2
$\leftarrow$ pos 4: feature 1, dist=1

Step 3: Query Selects Row of S

The query's one-hot vector selects row 2 of the scores matrix:

$$X_q \cdot S = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 & 3 & 1 & -1 & -3 \\ 3 & 4 & 3 & 1 & -1 \\ 1 & 3 & 4 & 3 & 1 \\ -1 & 1 & 3 & 4 & 3 \\ -3 & -1 & 1 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 & 3 & 1 \end{bmatrix}$$

This row encodes: "how much should I attend to each *feature*?"

Step 4: Assign Scores to Context Positions

Each context item receives a score based on its feature index:

$$\text{scores} = (X_q \cdot S) \cdot X_c^\top = \begin{bmatrix} 1 & 3 & 4 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 & \mathbf{3} \end{bmatrix}$$

pos 0: feature 0 $\leftarrow S[2,0] = 1$
pos 1: feature 4 $\leftarrow S[2,4] = 1$
pos 2: feature 0 $\leftarrow S[2,0] = 1$
pos 3: feature 4 $\leftarrow S[2,4] = 1$
pos 4: feature 1 $\leftarrow S[2,1] = 3$ ✓

Step 5: Softmax Selects Closest

Applying softmax concentrates probability on the highest score:

$$\text{softmax}\big([1,1,1,1,3]\big) \approx \begin{bmatrix} 0.09 & 0.09 & 0.09 & 0.09 & \mathbf{0.65} \end{bmatrix}$$

The model attends most strongly to **position 4**, which contains the context item with the closest feature (feature 1, distance 1 from query feature 2).

Step 6: Compute Values

With $\gamma = 0$ and learned $W_v = I$ (identity), the values are simply the context vectors:

$$V = \big[(1-\gamma)X_c + \gamma I\big]W_v = X_c \cdot I = X_c$$

Step 7: Weighted Sum of Values

The output is the attention-weighted sum of values:

$$\text{output} = \text{softmax}(\text{scores}) \cdot X_c = \begin{bmatrix} 0.09 & 0.09 & 0.09 & 0.09 & 0.65 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0.18 & \mathbf{0.65} & 0 & 0 & 0.18 \end{bmatrix}$$

Step 8: Prediction

Taking arg max of the output vector:

$$\hat{y} = \arg\max \begin{bmatrix} 0.18 & \mathbf{0.65} & 0 & 0 & 0.18 \end{bmatrix} = 1$$

The model correctly predicts **feature index 1**, which is the feature of the closest context item.

Key insight: The banded structure of S ensures that context items with features closer to the query's feature receive higher attention weights, enabling the model to solve the feature-identification task.